Style Before Memory: When Benchmark Exposure Starts to Inflate VLM Scores

Minho Chung mchung45@asu.edu
Yoorhim Cho yourmejo@asu.edu
Kaustav Chanda kchanda3@asu.edu
Injoh Tanwani itanwani@asu.edu

Abstract

Existing VLM contamination studies detect benchmark exposure, typically on models deliberately fine-tuned on evaluation items, leaving the effects of contamination dose and exposed modality largely unexplored. Contamination gains are also often read as memorization, although a model may instead adapt to a benchmark's caption style. We hypothesize that such style adaptation emerges at lower exposure levels than memorization of individual references. To test this, we run a controlled captioning study on NoCaps that varies repetition and exposed modality, comparing seen with unseen images and training with held-out references. This design lets us identify when exposure begins to inflate VLM scores and whether the gains reflect style or memorized references. We also test whether current detectors remain reliable at realistic contamination levels.

1 Introduction & Problem Statement

Benchmark scores for vision-language models (VLMs) are getting harder to take at face value. Benchmark images and captions are public, and many of them end up in pretraining or instruction-tuning data. Since training corpora are only partly disclosed, it is rarely possible to check how much overlap a given model has had (Song et al., 2025). Recent work has started to detect this exposure using perturbed inputs (Park et al., 2026), swapped tasks (Liu and Zhang, 2025), or masked captions (Song et al., 2025). These detectors are usually validated on models fine-tuned heavily on the test items, so they say little about weaker exposure, or about whether it was the image or the text that the model picked up.

Knowing that a model was exposed also does not tell us why its score went up. Take captioning. Benchmark captions tend to be short, single-sentence descriptions written in a fairly fixed register, quite unlike the long paragraphs an instruction-tuned VLM writes by default. After seeing enough of these captions, a model can learn to write the same way, and its CIDEr can rise on every image, including images it was never trained on. The score is inflated, but no particular reference has been memorized. We call this *style adaptation* and contrast it with *reference memorization*, where the model reproduces the specific caption it was trained on for a specific image. Our hypothesis is that style adaptation needs far fewer repetitions. Every caption carries the same style, so each exposure reinforces it; the link between one image and one sentence is reinforced only when that item comes up again (Figure 1, bottom).

NoCaps (Agrawal et al., 2019) gives us a way to tell the two apart (Figure 1, bottom). Each validation image has ten human-written captions, so we can train on one and keep the rest for scoring. We expose half of the images during additional fine-tuning, varying how many times each item is repeated and what the model actually sees: only the image, only the caption, both with captions shuffled across images, or the correct pair.

2 Research Objectives & Questions

We train controlled VLM checkpoints that vary repetition level and exposure modality under the same training and optimization budget. The study has three objectives:

- Track caption style, caption quality, and reference reproduction as exposure is repeated, comparing exposed and unexposed images.
- Separate the effects of image exposure, caption exposure, and image-caption pairing.
- Test existing contamination detectors under the same conditions, including false positives on negative controls.

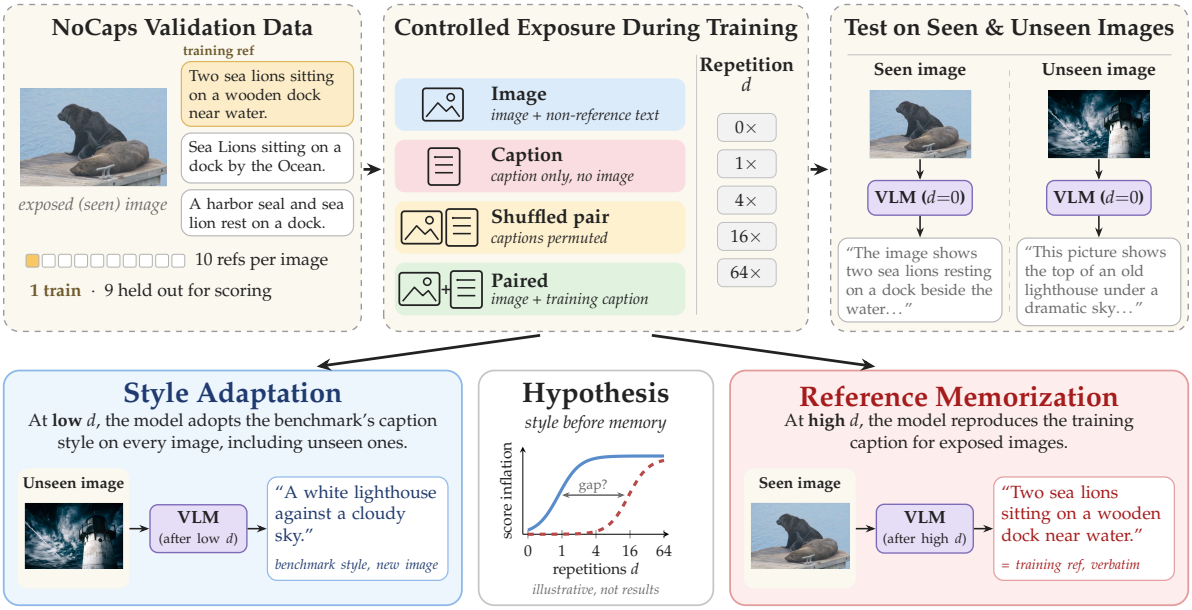


Figure 1: Overview of the study. Half of the NoCaps validation images are exposed during fine-tuning, using one of their ten captions; the rest stay unseen. We vary the repetition level d and what is exposed. At low d, we expect style adaptation: benchmark-style captions on every image, including unseen ones. At high d, we expect reference memorization: the training caption reproduced on seen images. Model outputs and curves are illustrative. Images from the NoCaps validation set (Open Images, CC BY 2.0).

RQ1. At what repetition level do style adaptation and reference reproduction become measurable, and how do they change with further exposure? We expect style adaptation to appear earlier because stylistic patterns are shared across many captions, whereas reproducing a particular reference may require repeated exposure to that item. We compare exposed and unexposed images, as well as training and held-out references, to determine whether gains are broadly shared or concentrated on specific images and reference wording. We also examine whether the changes are monotonic.

RQ2. How do the exposed modality and image-caption pairing affect these changes? We compare image-only, caption-only, shuffled-pair, and paired. We expect paired exposure to produce the strongest item-specific gains and reference reproduction. The shuffled condition separates co-exposure from correct image-caption alignment, while comparison with the unimodal conditions tests whether the paired effect exceeds their combined effects.

3 Methodology

3.1 Model and Benchmark

The primary model is LLaVA-1.5-7B (Liu et al., 2024), adapted with LoRA (Hu et al., 2022). We verify from documented data sources that the evaluation split was not intentionally used in training; otherwise, we use a smaller 1B–3B open VLM with better-documented training data. Selected low- and high-exposure conditions also use broader fine-tuning to test sensitivity to update scope.

We use NoCaps as the primary benchmark because its multiple human-written references per image enable separate comparison against training and held-out captions. Half of the evaluation images are randomly selected for controlled exposure, stratified by in-, near-, and out-of-domain subsets, while the rest form the unexposed set. For each exposed image, one reference caption is used for training and the others are held out for evaluation. We also construct a class-disjoint split to assess vocabulary transfer.

3.2 Controlled Exposure Design

We separately control exposure breadth, repetition, and dilution. The exposed fraction is fixed at 50%, and repetition level d is varied over $0, 1, 4, 16, 64$. We use two dilution levels: the standard instruction mixture and a setting with ten times more non-exposed examples per exposed example. Each condition is repeated with at least two random seeds.

For each nonzero d, the same items are used across four exposure conditions (Figure 1):

- **Image:** the benchmark image is paired with a matched prompt and a non-reference target.
- **Caption:** the selected reference caption appears in a text-only example without its image.
- **Shuffled pair:** benchmark images and captions are both exposed, but captions are permuted across images.
- **Paired:** each benchmark image is paired with its selected reference caption under the captioning prompt.

We use two controls. The zero-exposure condition provides a common baseline, while the same-distribution control replaces exposed evaluation items with NoCaps-style examples outside the evaluation set, matched by example count and caption length. Exposed and control examples replace ordinary instruction examples so that token and optimization budgets remain fixed.

Selected repetition levels also include cropped and re-encoded images to test near-duplicate exposure. The full grid contains 68 primary runs across repetition, exposure, dilution, and seed settings, before same-distribution controls. We first pilot $d \in \{0, 4, 64\}$ under standard dilution before expanding to the full grid.

3.3 Evaluation and Analysis

Let $f_{d,c}$ denote the model trained at repetition level d under exposure condition c. Given image x_i and a fixed captioning prompt p_{cap}, the generated caption is

$$\hat{y}_{i,d,c} = f_{d,c}(x_i, p_{\text{cap}}). \tag{1}$$

Models are evaluated at temperatures 0 and 1.0. The primary metrics are CIDEr (Vedantam et al., 2015) and SPICE (Anderson et al., 2016). For group $g \in \{\text{seen}, \text{unseen}\}$, let $\mathcal{I}_g$ be its image set and $\mathcal{R}_i$ the references for image i. The group score is

$$S^g_{d,c} = \frac{1}{|\mathcal{I}_g|} \sum_{i \in \mathcal{I}_g} m(\hat{y}_{i,d,c}, \mathcal{R}_i)\,. \tag{2}$$

The same metric implementation and reference configuration are used across conditions. For exposed images, we score training and held-out references separately and measure exact match and longest shared word n-gram for reference reproduction.

RQ1: Repeated exposure. Across repetition levels, we track caption length and vocabulary overlap as indicators of style change, and exact match and shared n-grams as measures of reference reproduction. Scores against training and held-out references, together with performance on unexposed images, indicate whether gains concentrate on exposed references or extend beyond them.

To isolate gains specific to exposed items, we use the matched zero-exposure baseline S^g_0 and compute

$$\Delta_{d,c} = \left(S^{\text{seen}}_{d,c} - S^{\text{seen}}_0\right) - \left(S^{\text{unseen}}_{d,c} - S^{\text{unseen}}_0\right). \tag{3}$$

We interpret $\Delta_{d,c}$ as an item-specific additional gain rather than direct evidence of memorization. The same-distribution control further tests whether similar gains arise without exposure to the evaluation items.

RQ2: Exposure modality and pairing. At each repetition level, we compare the Image, Caption, Shuffled, and Paired conditions. Image and Caption estimate unimodal effects, while Shuffled versus Paired tests the contribution of correct image-caption correspondence. We also measure whether the paired effect exceeds the sum of the unimodal effects:

$$\Gamma_d = \Delta_{d,\text{pair}} - (\Delta_{d,\text{image}} + \Delta_{d,\text{caption}})\,. \tag{4}$$

A positive Γ_d indicates that the item-specific gain under Paired exposure exceeds the combined Image and Caption gains; it is treated as a comparative effect, not a direct memorization measure.

Results are also compared across NoCaps domains and between class-disjoint and random splits. As a secondary analysis, we test three detectors on the same checkpoints: DCQ (Golchin and Surdeanu, 2025), which asks the model to pick the original caption among word-level perturbations given the image; CDD (Dong et al., 2024), which measures how peaked the captions sampled at $T{=}1$ are; and KDS (Choi et al., 2025), which compares embedding similarity before and after fine-tuning. For DCQ and CDD, we report AUROC for separating exposed from unexposed images. KDS gives a dataset-level score, so we check whether it tracks the true repetition level. False positives are measured on the zero-exposure and same-distribution controls.

4 Project Plan & Timeline

The project proceeds in five phases (Table 1). Phase 2 is a go/no-go point: the pilot verifies the pipeline and measures per-run compute before the full grid. If compute is insufficient, we first drop $10\times$ dilution at $d{=}64$, then reduce the number of exposed images. Phase 5 analyses are optional.

Ph.	Tasks	Weeks
1	Build exposure splits, conditions, and controls; evaluation pipeline; zero-exposure baseline	2
2	Pilot ($d \in \{0, 4, 64\}$); measure compute; finalize grid	1.5
3	Train full grid and controls	3
4	RQ1/RQ2 analyses; detector evaluation	2
5	Optional analyses; final report	1.5

Table 1: Project timeline.

5 Expected Contributions & Impact

This project provides a controlled analysis of how benchmark exposure affects VLM evaluation across repetition levels and exposure modalities. By varying exposure frequency and comparing modality and pairing conditions, the study characterizes when contamination begins to affect free-form captioning scores and how different forms of exposure contribute to these changes. If low-dose exposure has little effect, the results help bound the practical risk of contamination for the studied setting; if measurable inflation occurs, they identify the conditions under which benchmark gains become less reliable. The resulting checkpoints and analysis protocol may also support future score correction and more transparent reporting of benchmark exposure during training.

6 Related Work

Multimodal contamination. Recent VLM studies detect contamination in deliberately fine-tuned models through semantic image perturbation (Park et al., 2026), task perturbation (Liu and Zhang, 2025), and tests that separate unimodal from cross-modal leakage (Song et al., 2025). These methods ask whether a model was exposed, typically after full fine-tuning; none varies the exposure dose or injects image-only leakage.

Controlled contamination and detection. In text, controlled injection shows that even a single replica of MATH test items lowers loss, while later training and higher sampling temperature weaken the effect (Schaeffer et al., 2026), and that joint source–target exposure inflates translation scores far more than one-sided exposure (Kocyigit et al., 2025). Spiking test items at known rates further enables calibrated score correction (Wei et al., 2026). Detectors such as DCQ (Golchin and Surdeanu, 2025), CDD (Dong et al., 2024), and KDS (Choi et al., 2025) have been validated mainly on text, and similarity-based signals can flag models with no plausible exposure (Xu et al., 2026); we test whether such detectors remain reliable under controlled caption dose and exposure modality.

None of these studies provides a controlled VLM dose–response analysis that separates image-only, caption-only, and paired exposure while distinguishing style adaptation from reference memorization; this is the gap our study addresses.

7 Potential Future Extensions

Future work can extend the analysis to other VLM architectures, benchmarks, and multimodal tasks to test whether the observed contamination patterns generalize beyond image captioning.

References

Harsh Agrawal, Karan Desai, Yufei Wang, Xinlei Chen, Rishabh Jain, Mark Johnson, Dhruv Batra, Devi Parikh, Stefan Lee, and Peter Anderson. 2019. NoCaps: Novel object captioning at scale. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8947–8956. IEEE.

Peter Anderson, Basura Fernando, Mark Johnson, and Stephen Gould. 2016. SPICE: Semantic propositional image caption evaluation. In *European conference on computer vision*, pages 382–398. Springer.

Hyeong Kyu Choi, Maxim Khanov, Hongxin Wei, and Yixuan Li. 2025. How contaminated is your benchmark? Measuring dataset leakage in large language models with kernel divergence. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pages 10666–10682.

Yihong Dong, Xue Jiang, Huanyu Liu, Zhi Jin, Bin Gu, Mengfei Yang, and Ge Li. 2024. Generalization or memorization: Data contamination and trustworthy evaluation for large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 12039–12050.

Shahriar Golchin and Mihai Surdeanu. 2025. Data contamination quiz: A tool to detect and estimate contamination in large language models. *Transactions of the Association for Computational Linguistics*.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*.

Muhammed Yusuf Kocyigit, Eleftheria Briakou, Daniel Deutsch, Jiaming Luo, Colin Cherry, and Markus Freitag. 2025. Overestimation in LLM evaluation: A controlled large-scale study on data contamination's impact on machine translation. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pages 31105–31132.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2024. Improved baselines with visual instruction tuning. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26286–26296. IEEE.

Ming Liu and Wensheng Zhang. 2025. Reasoning multimodal large language model: Data contamination and dynamic evaluation. *arXiv preprint arXiv:2506.07202*.

Jaden Park, Mu Cai, Feng Yao, Jingbo Shang, Soochahn Lee, and Yong Jae Lee. 2026. Contamination detection for VLMs using multi-modal semantic perturbation. In *International Conference on Learning Representations (ICLR)*.

Rylan Schaeffer, Joshua Kazdan, Baber Abbasi, Ken Ziyu Liu, Brando Miranda, Ahmed Ahmed, Fazl Barez, Abhay Puri, Stella Biderman, Niloofar Mireshghallah, and Sanmi Koyejo. 2026. Quantifying the effect of test set contamination on generative evaluations. *arXiv preprint arXiv:2601.04301*.

Dingjie Song, Sicheng Lai, Mingxuan Wang, Shunian Chen, Lichao Sun, and Benyou Wang. 2025. Both text and images leaked! A systematic analysis of data contamination in multimodal LLM. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 10527–10542.

Ramakrishna Vedantam, C Lawrence Zitnick, and Devi Parikh. 2015. CIDEr: Consensus-based image description evaluation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4566–4575.

Johnny Tian-Zheng Wei, Jerry Li, Ameya Godbole, and Robin Jia. 2026. Correcting test set contamination by spiking the training data. *arXiv preprint arXiv:2605.24818*.

Bruce Changlong Xu, Lan Wu, and Alexander Ryu. 2026. A controlled audit of pretraining contamination in public medical vision-language benchmarks. *arXiv preprint arXiv:2606.10066*.